

Sample Quiz 1

DATA 412/612

Write your answers in an R script. You may use R and RStudio. You may **not** use the internet or your notes. Do **not** use `dplyr`, `filter()`, or `select()` on this quiz. Use Base R, including `subset()` and `function()`.

1. `subset()`

This data frame is already in your environment:

```
offers <- data.frame(
  firm    = c("Helix", "Nimble", "Helix", "Orbit", "Nimble", "Orbit", "Helix"),
  role    = c("intern", "intern", "analyst", "intern", "analyst", "analyst", "intern"),
  salary  = c(22, 19, NA, 21, 28, 27, 24),
  remote  = c(TRUE, FALSE, TRUE, TRUE, FALSE, TRUE, TRUE)
)
```

(a) Using `subset()` only (not `[]` and not `dplyr`), keep the **intern** rows whose **salary** is **strictly greater than the mean salary of interns**. Compute that intern-only mean with missing values removed. Drop the **role** column from what you return.

(b) Using `subset()` on the **salary vector** (not on the whole data frame), return every salary that is at least 22.

2. Writing a function

Write a function named `pair_mean_call` in an R script. It takes one argument, a data frame `df`.

- If `df` has fewer than 2 columns, return `"thin"`.
- If `df` has more than 2 columns, return `"wide"`.
- If the first column or the second column is not numeric, return `FALSE`.
- If either of those two columns contains any `NA`, return `"missing"`.
- Using a pipe, compute the mean of the **finite** values in column 1 and the mean of the **finite** values in column 2. (`Inf`, `-Inf`, and `NaN` are not finite. They are also not `NA`; do not treat them as missing for the previous bullet.)
- Return 1 if the first mean is greater than or equal to the second mean. Otherwise return the **name** of the second column.

Run your function on each of these inputs:

```
data.frame(u = 1:2)
data.frame(u = 1:2, v = c("a", "b"))
data.frame(u = c(1, NA), v = c(2, 3))
data.frame(u = c(1, Inf, 3), v = c(10, 10, 10))
data.frame(low = c(2, 4), high = c(1, 1))
data.frame(low = c(1, 1), high = c(2, 4))
```